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Corrosion Detection in High-Temperature Oil Pipelines Using Machine Learning

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Abstract

Ensuring the reliability and integrity of oil and gas pipelines relies heavily on the early detection of internal corrosion threats. This study introduces a machine learning (ML) based predictive model designed to analyze real-time operational data and anticipate the development of internal corrosion. The approach employs a feed-forward multilayer neural network model to predict corrosion rates specific to high-temperature pipeline environments. Results demonstrate that the model can accurately predict internal corrosion threats with an accuracy of 96.7% thereby validating its efficacy and demonstrating its potential for practical implementation in real-world scenarios. Integrating this approach with real-time operational monitoring offers oil and gas operators a practical and data-driven tool to enhance safety, reduce costs, and improve the overall reliability of pipeline operations.

Keywords: Corrosion Detection, Machine Learning, Pipeline, Oil and Gas, Neural Networks, Prediction

Introduction

Corrosion is the gradual degradation of materials, particularly metals, caused by chemical reactions with the environment, typically involving oxygen and water [1]. In the oil and gas industry, it poses a major threat to the safety and reliability of infrastructure such as pipelines, wells, and processing facilities. Its progression depends on factors like material selection, heat treatment, and environmental conditions. Corrosion can lead to structural failures, production downtime, and environmental harm, with costs extending beyond component replacement to include inspection and labor expenses. Corrosion itself can manifest in various forms such as general corrosion, pitting, crevice corrosion, and other localized or flow-induced types in which experienced inspectors can distinguish based on their patterns and affected areas [2]. Given its broad impact on operational performance and asset longevity, developing effective methods to monitor and predict corrosion has become increasingly vital to ensure the safety and efficiency of the oil and gas operations.

Overview on the paper structure

The paper is organized into several sections, each focusing on a specific aspect of corrosion detection in high-temperature oil pipelines using machine learning. The introduction provides an overview of the importance of corrosion detection and the limitations of traditional methods. The literature review section discusses the current state of research in corrosion detection, including traditional methods and machine learning approaches. The methodology section describes the dataset collection, data preprocessing, and feature engineering steps, as well as the model design and architecture. The results and evaluation section presents the performance of the models, and the conclusion and future direction section summarizes the findings and suggests potential avenues for future research.

Problem Definition & Objective

The problem addressed in this paper is the detection of internal corrosion in high-temperature oil pipelines. The objective of this research is to develop a machine learning-based predictive framework that can analyze real-time operational data and anticipate the development of internal corrosion.

The specific research objectives are:

- To develop a machine learning model that can accurately predict internal corrosion in high-temperature oil pipelines.
- To evaluate the performance of different machine learning algorithms, including random forests, extreme gradient boosting, and artificial neural networks.
- To investigate the effect of class imbalance on the performance of the models and to develop strategies to mitigate this issue.
- To demonstrate the potential of machine learning techniques in enhancing the safety and reliability of oil and gas operations.

Literature Review

This section reviews traditional non-destructive testing (NDT) methods as well as recent advancements using machine learning and deep learning techniques.

Traditional Methods for Corrosion Detection

The oil and gas industry has long utilized non-destructive testing (NDT) methods to detect corrosion, as these techniques allow for the evaluation of material integrity without causing damage, thereby ensuring safe and reliable operations [3]. This section will explore some of the widely used NDT methods reviewing their capabilities and limitations.

In-Line Inspection Method. One commonly employed method is in-line inspection (ILI), which utilizes smart pigs which are robotic tools inserted into pipelines to detect corrosion, cracks, and metal loss using advanced signal processing techniques. However, despite its enhanced capabilities, ILI is hindered by limitations such as measurement errors, high costs, and incompatibility with non-piggable pipelines, which can lead to increased costs and reduced accuracy due to noisy data and the need for multiple runs [4].

Acoustic Emission Method (AE). Acoustic Emission (AE) is another NDT method which offers real-time corrosion detection by capturing high-frequency sound waves emitted during material deformation [5]. Nevertheless, its effectiveness is compromised by environmental noise, particularly since corrosion signals typically occur at low frequencies, making it challenging to distinguish between actual corrosion activity and background interference [6].

Magnetic Particle Inspection Method. Magnetic Particle Inspection (MPI) is another widely used approach that leverages magnetic fields and particles to identify surface defects [7]. However, its

applicability is restricted to ferromagnetic materials, excluding non-ferromagnetic metals such as aluminum, copper, and austenitic stainless steel [8].

Radiographic Testing Method. Radiographic testing is the use of X-rays to find any internal defects of an object. This method is used in many industries such as in oil and gas, and is useful in testing for corrosion and in weld inspection [9]. However, as with all processes and methodologies, radiographic inspection has also some drawbacks including high costs, inability to assess surface corrosion or estimate corrosion depth, and the health risks associated with radiation exposure [10].

In conclusion, all of these NDT methods have been extensively utilized for corrosion detection in the oil and gas industry. However, each method has its own limitations, which emphasizes the need for continued research of safer and more effective corrosion detection methods. By integrating ML models into existing detection techniques, they can significantly enhance precision, optimize workflows, and automate inspections [9].

Applications of Machine Learning and Deep Learning Techniques in Corrosion Detection

Corrosion detection represents a challenging problem area. If not performed accurately, corrosion detection and monitoring could lead to catastrophic failures and financial losses. Recent advances in machine learning and deep learning have been evaluated to predict and detect corrosion in real-time. This section focuses on summarizing the current state of research using various approaches, highlighting the strengths and weaknesses of different algorithms and models.

Random Forest Approach. In a study published by NPJ Materials Degradation, a thorough review of machine learning methods for predicting corrosion was performed from a data-oriented standpoint. They demonstrated that the use Random Forest (RF) was useful for predicting corrosion. They also reported that, RF improved prediction accuracy with increasing training dataset size [11]. Similarly, a study by [12] was able to compare the predictive performance of several machine learning techniques, including, k-nearest neighbor, multilayer perceptron, random forest, and support vector regression models. The random forest algorithm consistently provided the best predictive performance, with strong accuracy particularly as training set size increased. In light of that, the application of RF in corrosion prediction becomes very important as such models assist in the identification and mitigation of possible corrosion threats thereby facilitating preemptive maintenance actions.

Deep Learning Approaches. As outlined in [13, 14], the comprehensive evaluation of various implemented neural network models for anomaly detection reveals the advanced capabilities of machine learning technologies towards improving corrosion detection accuracy and dependability. Convolutional Neural Networks (CNNs) have emerged as a particularly effective approach for image-based corrosion detection. The application of some variants of Regional CNN and ensemble models capture the most remarkable performance in pinpointing specific corrosion areas. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are powerful in analyzing sequential data and are able to accurately forecast future corrosion states based on historical corrosion data, thus excelling in time-series analysis. In addition, Generative Adversarial Networks (GANs) have been applied in pipeline integrity management for data augmentation and improving the detection of anomalies. Overall, the reviewed studies suggest that deep learning techniques offer considerable improvement in corrosion detection and management.

Hybrid Approaches. Wu et al. [15] presented a new and powerful technique for predicting corrosion through dual-driven data and knowledge neural network (DKNN). This technique combines data-driven learning, which uses past corrosion data and relevant environmental conditions, with knowledge-driven learning which uses the expert knowledge and the theory of the processes that causes corrosion. By combining both approaches, the model achieves remarkable accuracy and reliability which makes it very useful for corrosion prediction. In another work, studies by [16, 17] concluded that ensemble methods

achieved higher prediction accuracy compared to a single method. These findings highlight the growing effectiveness of hybrid and ensemble approaches in improving the accuracy and reliability of corrosion prediction.

Challenges of Corrosion Detection

Despite ongoing advancements in sensing technologies and the integration of Machine Learning (ML) and Deep Learning (DL) methods in corrosion detection systems, several technical and practical challenges continue to occur. These obstacles impact system reliability, accuracy, and scalability of real-time monitoring. Key issues relate to environmental conditions, sensor limitations, data processing difficulties, and the limitations of current ML/DL models in real-world deployment.

A comprehensive review conducted by Rajendran and Deivalakshmi [13] revealed that deep learning approaches tend to outperform traditional methods. Similarly, Almadhe et al.'s study [18] highlighted the capabilities of models such as Artificial Neural Networks, Random Forests, and Convolutional Neural Networks in detecting corrosion and optimizing maintenance. However, as we delve deeper into the applications of these technologies, it becomes clear that several challenges must be overcome. For instance, the sheer volume of data required to train the models, and the lack of standardized datasets makes it difficult to benchmark their performance. Ultimately, addressing these challenges will be crucial to unlocking the full potential of machine learning and deep learning in the context of pipeline integrity management.

Methodology

The methodology employed in this study involves a comprehensive approach to develop a machine learning-based predictive framework for corrosion detection in high-temperature oil pipelines. The process comprises several stages, including dataset collection, data preprocessing and feature engineering, model design and architecture, and model evaluation.

Dataset Collection

The dataset used in this study was collected from a real-world system operating in the Middle East. It contains approximately 9,000 records, each representing an event or condition recorded in real operational environments and classified into severity levels from 1 to 4, corresponding to increasing corrosion severity.

The dataset captures a broad spectrum of environmental and operational variables typical of the Middle Eastern region, including high ambient temperatures and elevated pressure levels, which are known to influence corrosion rates. The dataset was carefully curated to ensure representativeness of the operating conditions of oil pipelines in the region, thereby providing a reliable foundation for analysis and model development.

Below, in Table 1, shows the list of key fields in the dataset, along with their descriptions:

Table 1—Dataset Fields and their Corresponding Descriptions

FIELD NAME	DESCRIPTION
MATERIAL	Type of material being tested for corrosion
CHROME	Chromium content in the material
TEMP	Temperature of the environment
PRESS	Pressure of the environment
H2O	Water content in the environment
O2	Oxygen content in the environment
SO2	Sulfur dioxide content in the environment
NO2	Nitrogen dioxide content in the environment
H2S	Hydrogen sulfide content in the environment
TIME	Duration of the corrosion test
CR	Corrosion rate of the material
SEVERITY	Severity of corrosion (1-4, with 1 being mild and 4 being severe)

Data Preprocessing and Feature Engineering

The data preprocessing pipeline involved multiple steps to prepare the dataset for modeling. Initially, data cleaning was performed to handle missing values and remove any inconsistent or noisy entries that could potentially bias the models for example, issues within the MATERIAL (categorical) and SEVERITY (target) features. Categorical features such as SEVERITY were encoded using appropriate transformation techniques, such as one-hot encoding, to convert them into a machine-readable format. Numerical features such as TEMP and PRESS were standardized using StandardScaler to ensure uniform feature scaling across different units and magnitudes, which helps to prevent feature dominance and improves model performance. The target labels, originally in the range of 1 to 4, were re-encoded into numerical classes from 0 to 3 to align with standard classification practices. Given the noticeable class imbalance in the dataset, particularly the underrepresentation of certain severity classes, the Synthetic Minority Oversampling Technique (SMOTE) was employed to artificially generate new instances of the minority classes by interpolating between existing samples. This technique effectively improved class balance and helped prevent model bias toward the majority class.

The presence of class imbalance in the datasets posed a significant challenge to achieving optimal classification performance, as it disproportionately affected the recall and F1-score metrics for the minority classes. In order to gain a deeper understanding of the underlying data characteristics, an exploratory data analysis was undertaken, which encompassed the use of visualization techniques and correlation matrix assessments. This comprehensive analysis aimed to uncover potential relationships and redundancies among the features, thereby providing valuable insights into the data structure and informing the development of effective classification models.

Model Design and Architecture

The study employed multiple machine learning models with architectures suited to the nature of the dataset. Random Forest and ExtraTrees were used as ensemble methods based on decision trees, which build multiple trees using bootstrap samples and random feature subsets, aggregating their predictions to improve generalization and reduce variance. XGBoost was used as a gradient boosting framework that builds additive trees sequentially, optimizing a regularized objective to prevent overfitting. The model parameters were tuned to balance model complexity and performance. A feed-forward multilayer perceptron architecture was used for the neural network, consisting of an input layer, two fully connected hidden layers, and an output layer with a softmax activation function to handle multi-class classification. Dropout layers were incorporated after each hidden layer to mitigate overfitting by randomly deactivating a fraction of neurons during training. The network was trained using the categorical cross-entropy loss function optimized via the Adam optimizer. Each model was configured and trained to accommodate the imbalanced multi-class classification problem effectively.

To evaluate the performance of various algorithms on imbalanced multi-class classification, several machine learning models were implemented and assessed. The models included Random Forest Classifier, Extreme Gradient Boosting (XGBoost), and Artificial Neural Network (Multilayer Perceptron), each with its unique strengths and weaknesses. Each model was configured with tailored hyperparameters and class weighting strategies to mitigate the effects of class imbalance. The neural network was developed using the Keras library with a TensorFlow backend, employing a multi-layer architecture suitable for structured data. The models were trained and validated using a 70/30 train-test split, ensuring that 70% of the data was used for training while the remaining 30% was reserved for testing to provide reliable performance estimates and reduce the risk of overfitting.

The study employed multiple machine learning models with architectures suited to the nature of the dataset:

1. Random Forest and ExtraTrees: Ensemble methods based on decision trees. These models build multiple trees using bootstrap samples and random feature subsets, aggregating their predictions to improve generalization and reduce variance.
2. XGBoost: This is a gradient boosting framework that builds additive trees sequentially, optimizing a regularized objective to prevent overfitting. The model parameters were tuned to balance model complexity and performance.
3. Neural Network: A feed-forward multilayer perceptron architecture was used, consisting of:
 - An input layer with neurons corresponding to the number of input features.
 - Two fully connected hidden layers with 64 and 32 units respectively, both utilizing ReLU activation functions to introduce non-linearity.
 - Dropout layers were incorporated after each hidden layer to mitigate overfitting by randomly deactivating a fraction of neurons during training.
 - An output layer with a softmax activation function to handle multi-class classification.

The feed-forward multilayer perceptron architecture is illustrated in Fig.1 below:

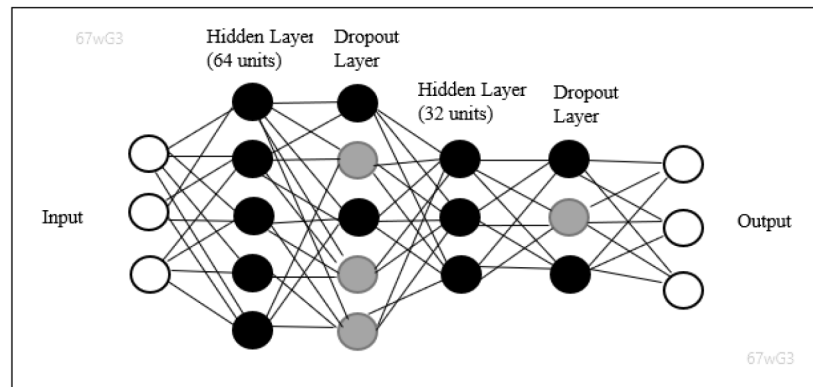


Figure 1—Feedforward Multilayer Perceptron Architecture

The network was trained using the categorical cross-entropy loss function optimized via the Adam optimizer. The depth and number of neurons were chosen to balance model capacity and overfitting risk, considering the dataset size (~9,000 samples). Dropout rates, learning rate, and batch size were tuned empirically using a validation set. Early stopping was employed to prevent overfitting during training. This clarifies the reasoning for the choice of neural network architecture and hyperparameters, and confirms that hyperparameter optimization was performed.

Memory Consumption

The memory footprint of the models varied, with neural networks generally requiring more memory than Random Forest and XGBoost models due to their complex architecture and larger number of parameters. Nevertheless, despite this increased memory consumption, the neural network model was ultimately chosen for its superior predictive performance, achieving the highest accuracy, precision, recall, and F1 score among the three models. The substantial gain in accuracy afforded by the neural network model outweighed the memory considerations, making it the preferred choice for this application, where high predictive precision is a primary objective.

Result and Evaluation

This section outlines the experiments conducted to evaluate different machine learning models for corrosion prediction. It details the experimental setup, model training, and performance analysis, followed by a comparison of results across key evaluation metrics.

Experimental Setup

All experiments were conducted using Python and leveraged several established libraries, including scikit-learn for traditional machine learning models, imbalanced-learn for handling class imbalance, XGBoost for gradient boosting, Keras with TensorFlow backend for neural networks, and CatBoost for additional gradient boosting experiments. The computational environment consisted of a Windows 10 workstation equipped with an Intel Core i7 processor and 16 GB of RAM. No GPU acceleration was utilized during training. To ensure reproducibility and consistency, the entire modeling pipeline, including data preprocessing, model training, and evaluation, was fully automated.

Training and Validation

To ensure a robust and unbiased assessment of each model's capabilities, we utilized a stratified split approach to divide our dataset into training and testing sets, with a train size of 70% and a test size of 30% of the dataset. This systematic split enabled us to maintain the same class distribution in both the training and testing sets, thereby mitigating potential biases and ensuring a more accurate evaluation of our models' generalizability.

Table 2—Training and inference times for Random Forest, XGBoost, and Neural Network models

Model	Training Time	Inference Time
RandomForest	2.1607 seconds	0.0219 seconds
XGBoost	129.46 seconds	0.0417 seconds
Neural Network	19.06 seconds	0.5576 seconds

The results reveals that the neural network model, chosen for its capacity to learn complex patterns and relationships within the data, exhibits a notable trade-off between training time and inference time. With a training time of 19.06 seconds, the neural network falls between the extremely fast RandomForest model and the computationally intensive XGBoost model, suggesting a moderate level of complexity in its architecture. However, its relatively high inference time of 0.5576 seconds raises concerns about its potential scalability and real-time applicability, particularly when compared to the much faster RandomForest and XGBoost models. Despite this, the neural network's ability to generalize and make accurate predictions, as evidenced by its performance on the test set, underscores its value as a robust and reliable model, warranting further optimization and refinement to improve its computational efficiency and responsiveness.

Analysis of the Results

Evaluation Metrics. Model performance was assessed using multiple evaluation metrics, including:

- Accuracy: The proportion of correctly classified instances out of all instances in the dataset.
- Precision: The ratio of true positives to the sum of true positives and false positives indicating the model's ability to avoid false alarms.
- Recall: The ratio of true positives to the sum of true positives and false negatives, reflecting the model's capacity to detect actual instances.

- Macro-averaged F1-score: The harmonic means of precision and recall, providing a balanced measure of both.

The macro-averaged F1-score was employed to provide a balance view of model performance across all classes. For each class in the set of classes, precision and recall are first computed as:

$$Precision = \frac{TP_i}{TP_i + FP_i}$$

$$Recall = \frac{TP_i}{TP_i + FN_i}$$

where TP_i denotes the number of true positives for class i , FP_i the number of false positives, and FN_i the number of false negatives. Using the previous formulas, the class-wise F1-score computed as follows:

$$F1_i = \frac{2 \cdot Precision_i \cdot Recall_i}{Precision_i + Recall_i}$$

Hence, the macro F1-score is obtained by taking the unweighted arithmetic mean of all class-wise F1-score:

$$F1_{macro} = \frac{1}{K} \sum_{i=1}^K F1_i$$

Model Performance Comparison. The following analysis compares the effectiveness of Random Forest, XGBoost, and Neural Network models based on the key metrics discussed above.

The observed results are summarized in the [Table 3](#) below:

Table 3—The Accuracy, Precision, Recall, and F1 scores for three different models

Model	Accuracy	Precision	Recall	F1
Random Forest	95.5%	95.5%	95.5%	95.5%
XGBoost	96.6%	95.5%	95.5%	96.6%
Neural Network	96.7%	96.7%	96.7%	96.7%

When comparing the performance of the three models, it is evident that the neural network model outperforms both the Random Forest and XGBoost models across all evaluated metrics. While Random Forest and XGBoost exhibit similar performance with approximately 95.5% precision and recall, the neural network achieves the highest accuracy, precision, recall, and F1-score of approximately 96.7%. This indicates that the neural network model not only excels in overall correctness but also maintains a balanced performance in both precision and recall, making it the most effective model among the three for this specific task.

The performance of the neural network model is a significant indicator of its potential to provide more accurate and reliable predictions, which can have a profound impact on maintenance and downtime reduction in the oil and gas industry. By leveraging the model's capabilities, organizations can enable proactive and precise interventions, thereby minimizing the likelihood of equipment failures and reducing the duration of unplanned downtime.

Confusion Matrix Analysis. In order to further evaluate the classification performance of the neural networks, a confusion matrix was generated as shown below:

The confusion matrix in [Figure 2](#) shows the neural network model's predictions compared to true class labels across the corrosion severity categories. The strong concentration of values along the diagonal indicates high accuracy, with examples such as class 0 having 654 correct predictions out of 675 and class 3

being perfectly classified. Off-diagonal values represent misclassifications, like the 29 instances of class 1 incorrectly predicted as class 0, but overall, the model effectively distinguishes between corrosion severity categories with minimal confusion.

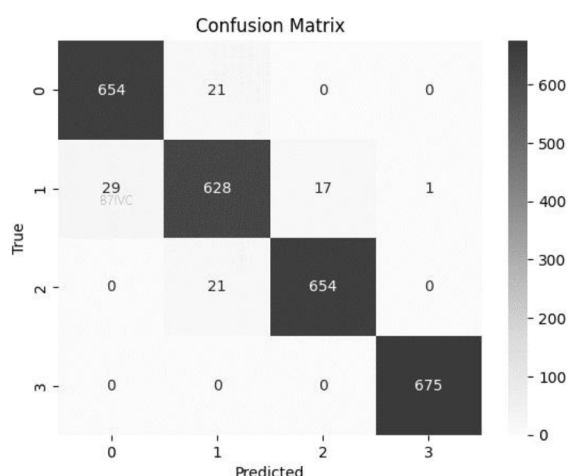


Figure 2—Confusion Matrix for the Neural Network

Neural Network Outperformance. The superior performance of the neural network in this study can be attributed to several architectural and functional advantages. Unlike Random Forest and XGBoost that are based on simpler decision trees, the neural network excels at capturing complex interactions through its layered structure and non-linear activation functions. This capability is particularly valuable in corrosion severity prediction, where environmental and material variables often influence each other in non-linear ways. Through a larger number of trainable parameters, the neural network also possesses a higher representational capacity, allowing it to model subtle variations in corrosion behavior more effectively. These characteristics contribute to its robustness and superior predictive accuracy in this application.

Conclusion and Future Direction

This study has demonstrated the effectiveness of machine learning models in detecting corrosion in high-temperature oil pipelines. The results have shown that carefully designed and trained models can achieve reliable performance, with the neural network approach yielding the most promising results. The findings of this research contribute to the development of more accurate and efficient corrosion detection systems, which can help reduce the risks and costs associated with pipeline failures.

Limitations

While the proposed approach has shown promising results, there are several limitations and potential risks that need to be considered. One of the primary concerns is the risk of false positives or false negatives in corrosion prediction. False positives can lead to unnecessary maintenance activities, resulting in increased costs and downtime, whereas false negatives can lead to undetected corrosion, potentially causing catastrophic failures. Furthermore, the performance of the model can be limited by the quality of the training data, which may not be representative of all possible operating conditions. The accuracy of the model can also be affected by various factors, including the quality of the sensor data, noise in the measurements, and the presence of outliers or anomalies. The approach used in this research assumes that the sensors and data acquisition systems are functioning correctly and providing accurate measurements. However, in practice, sensor failures, calibration issues, or data transmission errors can occur, which can compromise the accuracy of the predictions.

Future Direction

Future research directions may include exploring additional machine learning architectures, such as ensemble methods and other deep learning models, to further improve classification performance. Moreover, as the dataset continues to grow in size, it will be essential to develop strategies for efficient batch processing, distributed computing, and parallelization to ensure that the models can handle large volumes of data without compromising performance. Ultimately, the integration of these advanced machine learning approaches with real-time monitoring systems can enable proactive maintenance and improve the overall safety and reliability of oil and gas operations

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